Class 34: Bands, Free Electron Approximation, Tight Binding Approximation

In the last two classes we have looked at the origin of energy bands in solids both pictorially, as well as analytically, using the Schrödinger wave equation. While the flat band diagrams that can be drawn corresponding to the band structure of the solid, do have some utility, the $E\ Vs\ k$ relationship is able to explain material phenomena that cannot be explained using the flat band diagrams. This capability of the $E\ Vs\ k$ diagram, is something we will explore in greater detail in the next class.

In this class we will look at the utility of band diagrams and also examine two different starting points from which band structure can be explained.

Consider the band structures shown in Figure 34.1 below:

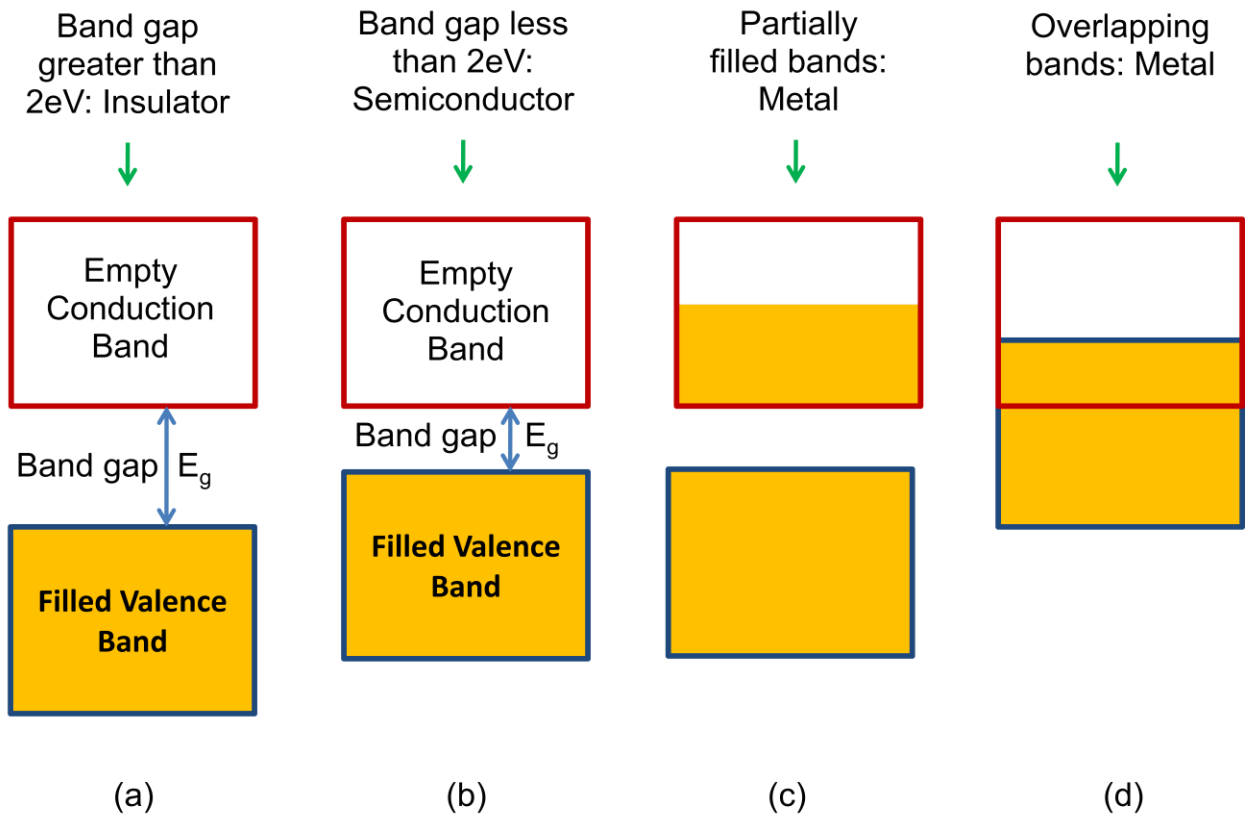


Figure 34.1: Band structure: (a) Band structure of an insulator; (b) Band structure of a semiconductor; (c) and (d) Band structures associated with a metal.

The band structure of an insulator consists of a fully filled valence band and an empty conduction band at higher energy. The band gap is typically greater than 2 eV. The band gap represents the minimum amount of energy that is required to enable electrons to move from the valence band to the conduction band. Once in the conduction band, the electrons are able to participate in the process of conduction.

The band structure of a semiconductor is similar to that of an insulator, except that the band gap is typically less than 2 eV.

Metallic systems have two possible band structures. Either they have a half filled band as the highest energy band, or they have an empty band and a full band overlapping in energy. In either case there are vacant energy levels immediately above the occupied energy levels and hence electrons are very easily able to find vacant states and participate in electronic conduction.

It is important to note that, by convention, the Fermi energy for insulators and intrinsic semiconductors, is assumed to be in the middle of the band gap, half way between the valence band and the conduction band. This is counterintuitive since by definition a band gap implies that there are no allowed energy levels in that range of energy. However this is the convention used since it helps explain material behavior. For extrinsic semiconductors, which are doped, the Fermi energy of the parent material moves up or down in the band gap, and now coincides with energy levels of the dopants.

In our analysis through all of the earlier classes, we have adopted a picture of the solid that says that there are ionic cores at fixed lattice locations and that there is a free electron gas enveloping these ionic cores. In other words we have assumed that the solid already exists and that the ionic cores are 'tightly bound' to their lattice locations while the **electrons are** '**free**' to run through the extent of the solid. This is called the '**Free electron approximation**'.

There is another approach to modeling materials which starts from a diametrically opposite position. In this approach, we do not have a solid to begin with. Instead the atoms are independent to begin with and are brought together to build the solid. All of the electrons are bound to their respective individual atoms to begin with. In this case the atoms are free to begin with while the **electrons are tightly bound to begin with**. In view of the focus on the electronic properties of the materials, this approach is referred to as the '**Tight binding approximation**' – highlighting the status of the electrons at the start of the model.

Figure 34.2 below shows how the Tight binding approximation builds the band structure of the solid.

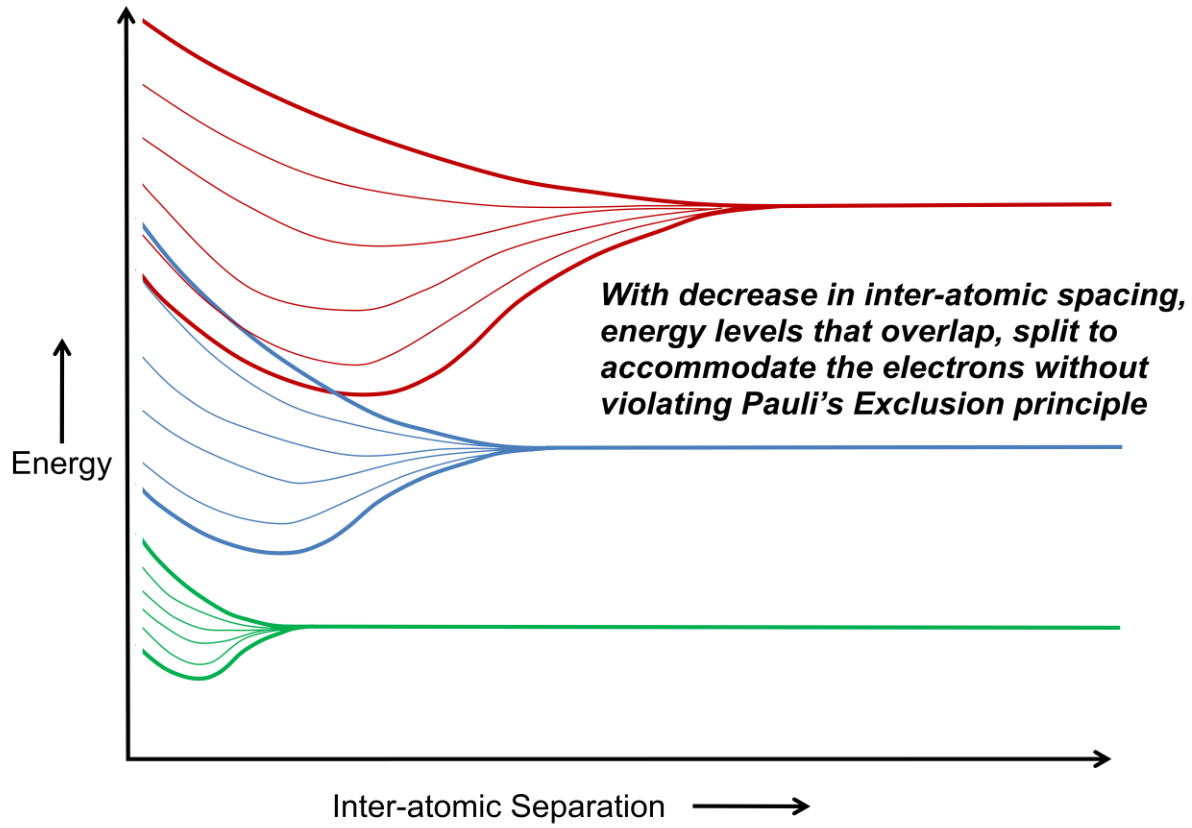
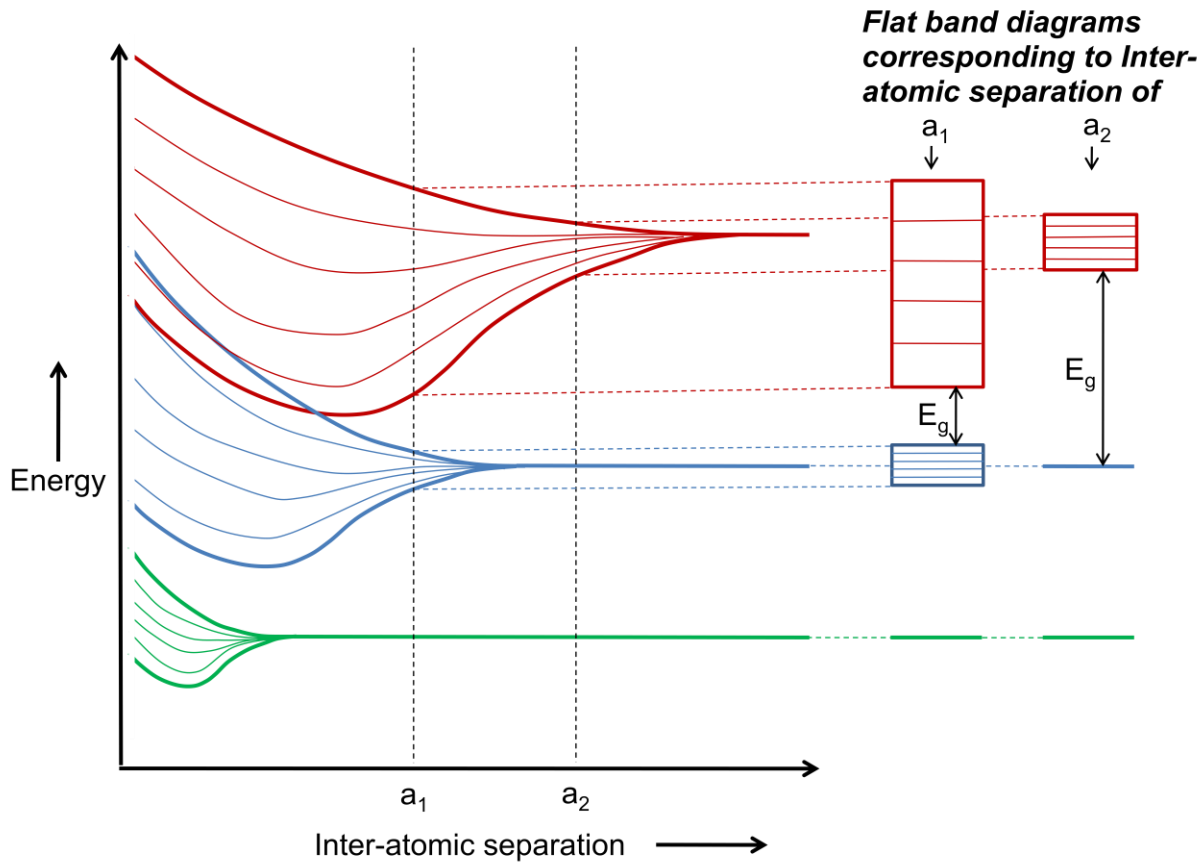


Figure 34.2: An illustration of the tight binding approximation to explain the properties of solids.

The tight binding approximation looks at the solid as follows: When the atoms are far apart, all of the bound electrons associated with each atom, have fixed energy levels that they can occupy. Assuming that we are building the solid using atoms of the same element, the energy levels occupied by the electrons in each atom will be identical. As we bring the atoms closer to each other to form the solid, as long as the interatomic separation is large, the electrons will still maintain their original energy levels. When the atoms get close enough that the outer shell electrons begin to overlap with each other, the energy levels of these outer shell electrons are forced to split into energy levels above and below the energy level of those electrons when they belonged to individual well separated atoms. This splitting of energy levels occurs because electrons obey the Pauli's exclusion principle (just as the energy levels could only hold limited numbers of electrons each in the Free electron approximation as well). Initially only the outer shell electrons overlap, therefore only their levels split and the inner shell electrons still maintain their individual atom based energy levels. If the interatomic separation keeps decreasing even further, progressively more of the inner shell electron levels will overlap and hence also split. At each energy level, the level will split to enough new energy levels so as to accommodate the electrons present in the original level for all of the atoms in the solid taken together. So, for example, if a hundred atoms come together, and there is one electron in the outer shell occupying one state, the solid will split this energy level to a hundred energy levels in the vicinity of this

original level to accommodate the hundred outer shell electrons corresponding to the combined solid.

Figure 34.2 above shows us what are all of the possibilities when we vary the interatomic spacing. In reality, a given material will have a specific interatomic spacing, which is the equilibrium spacing of that material under the experimental conditions that it experiences. Corresponding to this equilibrium spacing, a specific band structure, consistent with the tight binding approximation, will be displayed by the material. Figure 34.3 below shows us the different flat band structures that a material might display based on the choice of the equilibrium interatomic spacing.



Animation of the above is shown in the next page

Figure 34.3: Different flat band structures corresponding to different interatomic spacings chosen in a material. The existence of the bands, the width of each of the bands, as well as the extent of the band gaps change based on our selection of the equilibrium interatomic spacing. In this figure it is assumed that six atoms are being brought together, therefore each individual level splits into six levels when the energy levels overlap.

The reason the information in Figure 34.3 above is of interest is that we can experimentally manipulate the equilibrium interatomic spacing observed in the material. We can reduce the spacing by applying pressure on the material, or we can increase the spacing by applying a

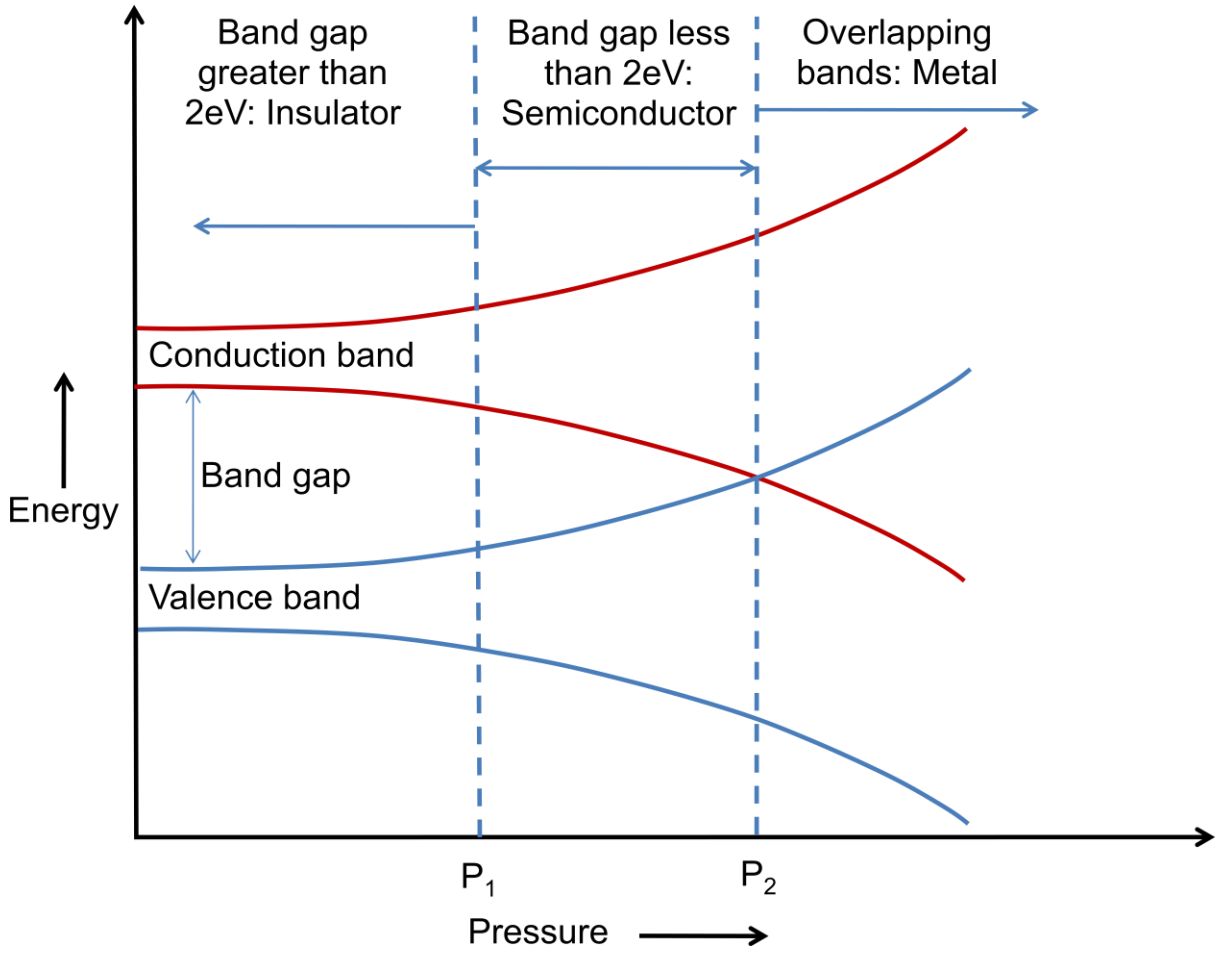
Animation of figure 34.3: Inter-atomic separation and corresponding band structure

Click here for animation on
Inter-atomic separation and corresponding band structure



tensile stress on the material which is less than the yield point of the material. Therefore, with the same material we can get different band structures, and that is of technological interest.

Since the interatomic spacing, and hence band structure can be changed by applying pressure, Figure 34.4 below shows the variation of the extent of the band and the extent of the band gap as a function of pressure.



Animation of the above is shown in the next page

Figure 34.4: Impact of pressure on the band structure of the material.

As seen from Figure 34.4 above, when the pressure applied is less than P_1, the material has a large band gap and is hence an insulator. When $P_1 < P < P_2$, the band gap is less than 2 eV and the material behaves like a semiconductor. When $P > P_2$, the bands overlap and the material behaves as a metal.

There are two aspects associated with Figure 34.4 above that we must note. The figure makes us aware that material properties are sensitive to pressure, and this includes electronic properties such as the band structure. Therefore, while the material may be synthesized and tested in the laboratory under one set of conditions, if it used in another set of conditions, its band structure and hence behavior, may be very different. While we tend to be more aware of the sensitivity of

Animation of figure 34.4: Pressure and corresponding band structure

Click here for animation on
Pressure and corresponding band structure



material properties to changes in temperature, we often take pressure being equal to 1 atmosphere, as granted. Depending on the end use, which could be in space, where the pressure is very low – essentially vacuum, or on other planets, where the pressure can be high, or in deep seas, where also the pressure can be very high, the pressure experienced by the material can definitely be significantly different from the ambient 1 atmosphere. So the effect of pressure should not be summarily ignored.

At the same time, we must also note that the sensitivity of material properties to changes in pressure is rather weak. In other words the pressure will typically have to change over several orders of magnitude to cause significant impact on material properties. For example, consistent with the Figure 34.4 above, Hydrogen gas is predicted to become a solid that is metallic at high pressures. But the pressure at which this change to a metallic hydrogen is expected to occur, is of the order of a million atmospheres, or six orders of magnitude higher pressure than that experienced under ambient conditions. Therefore, we are often not terribly in error in ignoring the effects of pressure, especially on solids.

We have therefore discussed two approaches: The free electron approximation and the tight binding approximation. In principle the two approaches must give the same results since they are modeling the same materials. As it turns out, largely they are consistent with each other and with the experimental data. However, in view of the starting points of these two approaches, the free electron approximation lends itself more easily to the treatment of metallic systems where the state of the material is consistent with the picture that the free electron approximation tries to address. The tight binding approximation is typically more consistent with the state of the material in the case of insulators, so it is better suited for modeling insulators.

We will close this class by noting that in the free electron approximation, Brillouin zone boundaries are an important factor in determining the band structure. Since the Brillouin zone boundaries occur at different distances along different directions in crystalline solids, the free electron approximation is able to indicate the cause for anisotropy in the properties of crystalline solids.

In the tight binding approximation, the interatomic spacing is a critical parameter in determining the band structure of the material. Since the interatomic spacing in crystalline solids varies based on the crystallographic direction chosen, the tight binding approximation is also able to explain the cause for anisotropy in the properties of crystalline solids.

Therefore both the approaches can explain anisotropy.

In the next class we will look at semiconductors, and examine how the theories we have developed so far help us understand the behavior of semiconductors.